Pediatric Narrow-Complex Tachycardia

PC-05 | Pediatric Cardiac

Quick flow

01 Narrow Tachycardia [EMT / AEMT / Paramedic]

Assess perfusion, mental status, BP, respiratory status, onset, P waves, R-R variability, and reversible causes.

02 Cardiopulmonary Compromise? [EMT / AEMT / Paramedic]

Hypotension • acutely altered mental status • signs of shock • sudden collapse. If yes, prepare synchronized cardioversion immediately.

03 Synchronized Cardioversion [Paramedic]

0.5-1 J/kg; if unsuccessful, increase to 2 J/kg. Confirm sync mode before every shock. Sedate only if immediately available and it does not delay treatment.

04 Stable + Regular SVT? [EMT / AEMT / Paramedic]

If regular narrow-complex SVT is likely, attempt vagal maneuvers first. Irregular rhythm or uncertain/wide-complex tachycardia: Medical Control and appropriate alternate pathway.

05 Adenosine [Paramedic]

If IV/IO is readily available: rapid push followed by rapid saline flush, then document rhythm response.

06 Reassess + Transport [EMT / AEMT / Paramedic]

Treat the cause, monitor continuously, obtain 12-lead when it does not delay care, and notify destination.

Overview

- Differentiate sinus tachycardia from SVT. Sinus tachycardia is treated by correcting its cause; SVT is typically abrupt, regular, and fixed-rate.
- Cardiopulmonary compromise from tachycardia requires prompt synchronized cardioversion. Sedation must never delay the shock.

Provider-Level Actions

Use a length- or weight-based pediatric resuscitation system and perform immediate lifesaving care without delaying oxygenation, ventilation, CPR, or defibrillation.

EMT

- Support airway and oxygenation as clinically indicated, obtain vital signs and rhythm strip when available, and identify cardiopulmonary compromise.
- Treat likely causes of sinus tachycardia - fever, pain, dehydration, hypoxia, sepsis, stimulant/toxin exposure, or anemia - within current standing orders.
- Keep the caregiver with the child when possible and prepare for immediate cardioversion if deterioration occurs.

AEMT

- Perform all EMT care plus IV/IO access and supportive treatment of the underlying cause within current standing orders.
- Do not delay transport or cardioversion for repeated vascular-access attempts.

Paramedic

- Perform all prior care plus synchronized cardioversion for cardiopulmonary compromise and vagal maneuvers/adenosine for stable, regular narrow-complex SVT.

- Use the most proximal practical IV/IO access for adenosine. Document rhythm before and after each intervention; obtain a 12-lead when stable without delaying treatment.
- For refractory stable SVT after vagal maneuvers and adenosine, contact Medical Control. Do not add antiarrhythmics not carried by the agency.

Indications

- Pediatric narrow-complex tachycardia (QRS at or below 0.09 seconds) with suspected SVT or serious symptoms.

Treatment steps

- Sinus tachycardia usually has visible P waves and variable R-R intervals; likely triggers must be addressed rather than treated with adenosine or cardioversion.
- SVT is usually abrupt and regular with absent/abnormal P waves; infant rate is often above 220/min and child rate above 180/min.
- Stable suspected SVT: vagal maneuvers first. If IV/IO access is readily available, give adenosine by rapid push and immediate saline flush.
- Do not give adenosine for an irregular rhythm, polymorphic rhythm, or uncertain/wide-complex tachycardia. Use PC-06 when the QRS is wide or diagnosis is uncertain.
- If the child becomes pulseless at any point, begin CPR and move immediately to PC-04 Pediatric Pulseless Arrest.

Medications

Adenosine

0.1 mg/kg rapid IV/IO push followed immediately by rapid saline flush; maximum 6 mg. If needed, repeat 0.2 mg/kg; maximum 12 mg. Only for regular narrow-complex SVT.

Midazolam

0.1-0.2 mg/kg IV/IO/IN for cardioversion; maximum single dose 2 mg, maximum total 5 mg

- Use only if immediately available; do not delay synchronized cardioversion. Verify against the current Claiborne pediatric medication reference.

Warnings / precautions

- Do not delay synchronized cardioversion for sedation in a child with cardiopulmonary compromise.
- Do not give adenosine for an irregular rhythm, polymorphic rhythm, or uncertain/wide-complex tachycardia.
- If pulseless at any time, begin CPR and move immediately to Pediatric Pulseless Arrest.
- Do not delay treatment for a 12-lead ECG or repeated vascular-access attempts.

Clinical pearls

- Adenosine effectiveness depends on a rapid bolus and immediate saline flush; rhythm documentation before and after administration is important.
- Tachycardia from fever, pain, dehydration, hypoxia, sepsis, toxin exposure, or anemia is usually sinus tachycardia and requires cause-directed treatment.

Related pathways / contacts

AR-05 Pediatric Airway - Open for airway and oxygenation support.

PC-04 Pediatric Pulseless Arrest - Open immediately if the child becomes pulseless.

PC-06 Pediatric Wide-Complex Tachycardia - Open for wide-complex or uncertain tachycardia.

Medication Reference - Open the current Claiborne medication reference.

Source and references

Claiborne County EMS approved protocol manual. | Current Tennessee EMS scope of practice and Claiborne County standing orders control for all provider permissions.

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